

Algorithmic Methods of Data Mining and Laboratory

Master's Degree in Data Science, Sapienza Università di Roma
Faculty of Information Engineering, Informatics and Statistics

Department of Computer Science

Author: Gianmaria Romano

Reference Bibliography

- C. Aggarwal: *Data Mining: The Textbook*
R. Zafarani, M. A. Abbasi, and H. Liu: *Social Media Mining: An Introduction*
J. Leskovec, A. Rajaraman, and J. Ullman: *Mining of Massive Datasets*
C. D. Manning, P. Raghavan, and H. Schütze: *Introduction to Information Retrieval*
J. Janssens: *Data Science at the Command Line*

Academic Year 2026/2027

Disclaimer

This document contains notes for the **Algorithmic Methods of Data Mining and Laboratory** course, given by **Professors Anagnostopolous and Siciliano** as part of the Master's Degree in Data Science at **Sapienza Università di Roma** during the **first** semester of Academic Year **2026/2027**.

Note: These notes may not cover all details of each lecture, so it is highly suggested to consult the Professor's official resources and materials.

Important: These notes are free to share and use, but please remember to credit me as the author and do not try to hide/remove this page.

Contacts

For further information or requests, feel free to contact me privately or reach out through my repository: <https://github.com/GianmariaRomano/Data-Science-Notes>.

License

These documents are distributed under the GNU General Public License 3.0, a form of copyleft intended for use on a manual, textbook or other documents. Material licensed under the current version of the license can be used for any purpose, as long as the use meets certain conditions:

- All previous authors of the work must be attributed.
- All changes to the work must be logged.
- All derivative works must be licensed under the same license.
- The full text of the license, unmodified invariant sections as defined by the author if any, and any other added warranty disclaimers (such as a general disclaimer alerting readers that the document may not be accurate for example) and copyright notices from previous versions must be maintained.
- Technical measures such as DRM may not be used to control or obstruct distribution or editing of the document.

Contents

1	Introduction to data mining	3
1.1	Finding a definition of data mining	3
1.2	Understanding data types	3

Chapter 1

Introduction to data mining

1.1 Finding a definition of data mining

While no unique definition exists yet, it is possible to view data mining as the use of efficient techniques for the analysis of very large collections of data and the extraction of useful and possibly unexpected patterns in data in order to make decisions across various areas.

Example: Example: Descriptive and predictive data mining

Descriptive data mining aims to identify patterns and relationships within data using unsupervised learning techniques that can be applied to recommendation systems, such as Netflix or Spotify's engines, to analyse patterns in user behaviour.

On the other hand, predictive data mining focuses on using past observations to predict a target variable through supervised learning techniques.

For instance, CERN's *A Large Ion Collided Experiment* analyses heavy-ion collision data to investigate the properties of quark-gluon plasma.

1.2 Understanding data types

Since data can come in different forms, it is necessary to organize it in order to extract meaningful information or patterns.

In particular, data can be classified into three main types:

- **Structured data:** Data is organized according to a well-defined schema that is enforced during storage.
- **Semi-structured data:** Data can be organized according to a structural schema, although this organization paradigm may not be enforced during storage.
- **Unstructured data:** Data does not follow a pre-defined structure, requiring a data cleaning procedure to store it as (semi-)structured data before use.

Example: Example: Different data types

Examples of data organization can be seen on the Infostud platform, which enforces strict organization through dedicated constraints on certain types of data, and medical records, which follow some structural patterns but might contain errors or anomalous data.

By contrast, social media platforms, such as Twitter or Reddit, provide clear examples of unstructured data that do not follow an organizational structure.